

TKS Project Report: Enhancing FormSubmit for Better Spam Protection

Research and Selection Process

I began my project by researching Y Combinator companies to find a suitable candidate for implementing and enhancing form functionality. During this research, I discovered BloomText, a promising platform for delivering private patient information to doctors. However, after reaching out to the founder and not receiving a response, I decided to proceed with FormSubmit, a platform I have been using for some time and felt confident about improving.

FormSubmit is a reputable company, having received \$86,000 in Series A funding, according to Crunchbase and LinkedIn. This validated the platform's potential and relevance for my project.

Features Added to FormSubmit

1. Dynamic Spam Filtering System:

- **Default Spam Word Database:** I created a comprehensive pre-defined list of common spam words that are automatically included in the blacklist. This helps in filtering out spam messages effectively.
- **Customizable User Blacklist:** This feature allows users to easily add, remove, and manage their own spam words, providing a more tailored spam protection experience.
- **JavaScript-Based Dynamic Blacklist Population:** A script was created to dynamically add blacklist words to the form, making it easier to manage and update spam words without manual intervention.

2. URL Filtering:

- **Ban Specific URLs:** I added an option to ban specific URLs from being included in form submissions, enhancing security and preventing spam links.
- **Complete URL Ban:** Users can now ban all URLs or allow only one specific URL in form submissions, providing more control over the content submitted through forms.

- Base64 Encoding for Allowed URLs: To enhance security, I implemented Base64 encoding for allowed URLs, preventing easy tampering and ensuring only the specified URLs are allowed.

3. Content Security Policy (CSP):

- Enhanced Security: I implemented a CSP to control where the website can load resources from, such as scripts, styles, and fonts. This prevents unauthorized scripts from executing on the site, protecting against Cross-Site Scripting (XSS) attacks.
- Controlled Resource Loading: The CSP ensures that only resources from trusted sources are loaded, reducing the risk of data breaches.
- Violation Monitoring: A reporting mechanism can be set up to monitor any CSP violations, allowing for real-time detection and response to potential security issues.

4. Anti-Spam Mechanisms:

- Form Submission Timing: I implemented a check to ensure the form is not submitted too quickly. If the form is submitted in less than 2 seconds from the time it was loaded, it is flagged as spam.
- Content Analysis: The message content is analyzed for spammy phrases using a predefined list of spam words. If any spam words are detected, the form submission is blocked.

5. Enhanced User Feedback:

- Popup Messages: Instead of using standard alert boxes, I created custom popup messages to inform users of validation errors. These popups are styled to match the form's aesthetics and provide a better user experience.

6. Tech Fluency Selector:

- User-Friendly Documentation: I added a tech fluency selector to the documentation that allows users to choose their comfort level with technology. Based on their selection, the documentation adjusts to provide explanations that match their understanding, ranging from basic to advanced levels.

7. Code Optimization and Maintenance:

- **Consolidated JavaScript:** All JavaScript functionality was consolidated into a single script within the HTML file, making it easier to manage and deploy.
- **Efficient URL Checking:** The URL validation logic was enhanced to handle URLs with or without the protocol (``http://`` or ``https://``), ensuring consistent validation.

8. Email Design Enhancements:

- **New Email Templates:** In addition to the security features, I designed new email templates that are more visually appealing, structured, and aligned with modern design standards. These templates offer improved readability and professionalism, enhancing the overall user experience.
- **Template Variations:** I created different email design variations such as "StreamlineNotify" and "Polished Precision," each offering unique benefits like clear visual hierarchy, better alignment, and aesthetic appeal.

Conclusion

Overall, I significantly improved the spam protection capabilities of FormSubmit, adding multiple layers of security and user-friendly features. These enhancements not only protect against spam but also provide users with a customizable and efficient form submission experience. The successful implementation of these features showcases the potential of FormSubmit as a reliable platform for handling form submissions securely.